

The NFSC Pedagogy: Narrative-Driven Cognitive Progression in Graduate Engineering Education

A Case Study from a Chinese Solid Waste Management Course

[Author names and affiliations removed for double-blind review. See separate Title Page.]

Abstract: Engineering graduate education faces a well-documented challenge: the disconnection between decontextualised textbook cases and the multidimensional complexity of authentic engineering problems. This paper presents the NFSC (Narrative–Framework–Scenario–Creation) pedagogy, a four-dimensional progressive teaching model developed and iterated over three academic years (2023–2025) in a graduate solid waste management course at a Chinese university. Drawing on Bruner's narrative theory, constructivist learning principles, situated learning theory, and outcome-based education, NFSC structures the learning process into a cognitive progression: narrative excerpts from documentary novels trigger affective engagement; a domain-specific Mismatch–Repair analytical framework provides diagnostic tools; three types of immersive scenarios (EIA debate, Nudge design, mock environmental court) temper application capabilities; and a three-track output system closes the learning loop. Using a practitioner case study design with thematic analysis of student outputs and classroom observations, we document three forms of cognitive change: shifts in causal attribution from intuitive moral judgment to framework-informed analysis, longitudinal growth in multi-dimensional framework usage, and diversification of creative outputs. The study's limitations—single course, small cohorts (~30 per cycle), absence of controlled comparison—are honestly acknowledged. We discuss transferability preconditions for other engineering disciplines and position this work as exploratory practitioner research that may inform larger-scale investigation.

Keywords: engineering education; graduate teaching; narrative pedagogy; scenario-based learning; constructive alignment; practitioner research

Introduction

Engineering graduate education worldwide faces a structural tension. Accreditation frameworks, from the Washington Accord (International Engineering Alliance, 2014) to ABET in the United States, increasingly demand competency-based outcomes—what graduates can **do**, not merely what they **know** (Crawley et al., 2014; Spady, 1994; Felder & Brent, 2003). Yet the dominant instructional paradigm in graduate engineering classrooms remains the lecture, which excels at efficient knowledge transmission but struggles to cultivate the multidimensional judgment required in authentic engineering practice (Biggs & Tang, 2011; Laurillard, 2012). This tension has been extensively documented in the engineering education literature. Sheppard et al. (2009) characterised the gap between "engineering science" and "engineering practice" as the central challenge of modern engineering education reform. Froyd et al. (2012) reviewed five major shifts in engineering education over the past century and concluded that despite curricular innovations, the fundamental **pedagogical** transformation at the individual course level remains incomplete.

This tension manifests acutely in cases used for teaching. Textbook cases are typically "clean problems": abstracted, decontextualised, and stripped of the political, social, and temporal complexity that characterises real engineering work (Lönngren et al., 2021). By contrast, authentic engineering practice involves "wicked problems" (Rittel & Webber, 1973): a landfill siting decision, for instance, involves not only geological and hydrological analysis but also community opposition, trans-jurisdictional accountability ambiguity, and temporal delays in environmental benefit realisation. The gap between textbook exemplars and professional reality is particularly pronounced in fields such as environmental engineering, where technical decisions are inextricably entangled with social, economic, and ethical dimensions (Jamieson & Lohmann, 2012).

Prior pedagogical frameworks have addressed aspects of this challenge. Problem-based learning (PBL) situates learning within authentic problems (Hmelo-Silver, 2004); the CDIO framework emphasises conceiving, designing, implementing, and operating real-world systems (Crawley et al., 2014); case-based teaching provides contextualised analytical practice (Herreid, 2011). However, these approaches typically assume that students already possess the analytical frameworks needed to diagnose presented problems, and that the motivational dimension—why students **care** about the problem—is either self-evident or incidental. Recent scholarship in the **European Journal of Engineering Education** has called for pedagogical designs that explicitly scaffold the motivational–cognitive transition (Edström & Kolmos, 2014; Lönngren et al., 2021), yet concrete course-level implementations remain scarce. The gap between pedagogical theory and classroom practice persists: instructors know **what** to aim for (competency-based outcomes) but lack proven **mechanisms** for achieving those outcomes within the temporal and resource constraints of a standard semester (Borrego & Bernhard, 2011; Case & Light, 2011).

The NFSC (Narrative–Framework–Scenario–Creation) pedagogy addresses this gap by explicitly building the motivational foundation **before** introducing analytical tools, and by structuring the entire learning experience as a cognitive progression rather than a collection of activities. Developed in a graduate course "Solid Waste Pollution Prevention and Control Technology" at [University], China, NFSC draws upon seven documentary network novels—non-

fiction prose totalling approximately two million characters—published by the course instructor as a narrative resource pool. Twenty-one teaching excerpts (coded S001–S021) selected from these novels provide authentic, emotionally textured accounts of real solid waste management conflicts across China.

This paper presents the theoretical grounding, instructional design, and eight-week implementation of NFSC, documenting cognitive changes observed across three teaching cycles (2023–2025). The study adopts a practitioner case study approach (Yin, 2018), using qualitative analysis of student outputs and classroom observations to provide "fieldnote" evidence of pedagogical effects. We make no causal claims—the inquiry is exploratory, positioning NFSC as a teaching innovation worthy of broader investigation rather than a proven intervention.

Our research questions are: (1) How does the NFSC progression influence students' causal attribution patterns when diagnosing solid waste management problems? (2) To what extent does framework usage develop longitudinally across the eight-week cycle? (3) What forms of creative output emerge from a three-track assessment system?

Literature Review

Engineering Education Reform and the Competency Gap

Engineering education has undergone substantial reform since the turn of the millennium, driven by accreditation standards that foreground competency-based outcomes. The Washington Accord (International Engineering Alliance, 2014), signed by over 20 national engineering bodies, established mutual recognition of accredited engineering programmes and catalysed a global shift toward outcomes-based accreditation. National adaptations such as ABET EC2000 in the United States (Felder & Brent, 2003) and the European EUR-ACE framework (Augusti, 2007) have institutionalised the expectation that graduates demonstrate not only technical knowledge but also design capability, ethical reasoning, communication skills, and lifelong learning capacity.

Despite these accreditation-level transformations, the dominant mode of instruction in graduate engineering classrooms remains the lecture (Graham, 2018; Froyd et al., 2012). Several comprehensive reviews have documented this gap between policy aspirations and classroom reality.

Sheppard et al. (2009) characterised the tension between "engineering science" and "engineering practice" as the central challenge of modern engineering education, arguing that the post-Sputnik emphasis on rigorous mathematical and scientific foundations—while successful in elevating the intellectual stature of engineering—had inadvertently marginalised the integrative, design-oriented, and context-sensitive dimensions of professional practice. Dym et al. (2005) made a parallel argument specifically for design thinking, contending that the analytical orientation of most engineering curricula systematically underprepares students for the synthetic and iterative nature of authentic design work.

The CDIO initiative (Crawley et al., 2014) operationalised competency-based expectations through a structured framework emphasising conceiving, designing, implementing, and operating. CDIO represented a significant advance over traditional lecture-based curricula by providing a coherent philosophy and a set of twelve standards for programme-level reform. Yet CDIO and similar frameworks are primarily *curriculum-level* interventions; they provide less guidance on *course-level* pedagogical strategies for individual instructors seeking to transform their own classrooms (Edström & Kolmos, 2014). Kolmos et al. (2016) noted that even within CDIO-adopting institutions, the translation from programme philosophy to daily teaching practice remains a persistent challenge, with many instructors defaulting to familiar lecture formats despite institutional commitment to active learning.

At the course level, the "last mile" problem persists: how to translate abstract competency statements into concrete, week-by-week instructional sequences. Constructive alignment theory (Biggs, 1996; Biggs & Tang, 2011) articulates the principle—learning activities and assessments must align with intended learning outcomes—but leaves the specific pedagogical *mechanisms* underspecified. How, precisely, does an instructor design activities that cultivate diagnostic thinking, systems reasoning, and ethical judgment within the constraints of a standard semester? This question has been recognised as a frontier challenge in engineering education research (Borrego & Bernhard, 2011; Case & Light, 2011), yet empirically grounded answers remain scarce.

Narrative Pedagogy in Higher Education

Narrative theory offers one answer to the mechanism question. Bruner (1986) distinguished between paradigmatic thinking (logical, propositional) and narrative thinking (story-based, contextual), arguing that humans organise experience and construct meaning primarily through narrative structures. This distinction has profound implications for pedagogy: if humans are "narrative creatures," then teaching that begins with abstract propositions—definitions, classifications, principles—may be working against, rather than with, the natural grain of human cognition.

In educational contexts, narrative pedagogy has been shown to enhance engagement, deepen comprehension, and facilitate the integration of abstract concepts with lived experience (Goodson & Gill, 2011; McDrury & Alterio, 2003). Narratives provide contextual scaffolds that enhance memory encoding and retrieval by embedding abstract concepts within emotionally resonant story structures (Immordino-Yang & Damasio, 2007). Pascarella and Terenzini (2005), in their comprehensive synthesis of over 2,500 studies on college impact, identified active, engaged pedagogical approaches as consistently associated with deeper learning outcomes.

In engineering education specifically, the potential of narrative remains underexplored. While case-based teaching is well-established (Herreid, 2011; Yin, 2018), cases are typically presented as *analytical objects*—"read the background, answer the questions"—rather than as *narrative experiences* that engage students affectively before engaging them analytically. This ordering matters. Kellam et al. (2018), in a narrative analysis of engineering students' emotional experiences, found that students' affective responses to course material profoundly shaped their cognitive engagement, yet these emotional dimensions are rarely deliberately designed into engineering pedagogy. Lönngren et al. (2020) made a compelling case for a research agenda on emotions in engineering education, arguing that the traditional model of learning as purely rational information processing—what might be termed a "cold cognition" approach—is both theoretically outdated and pedagogically limiting. Strobel et al. (2013) documented how practicing engineers themselves describe their professional motivation in deeply narrative and empathetic terms, suggesting a disconnect between how engineering is taught (as decontextualised problem-solving) and how it is experienced (as human-centred narrative practice).

The NFSC pedagogy directly addresses this affective-cognitive gap. By placing narrative engagement **before** analytical framework introduction, NFSC inverts the typical case-teaching sequence. Students first encounter a problem through the lived experience of characters in a story—feeling the land devaluation, the contaminated well water, the failed apricot trees—before being given the analytical vocabulary to name what they have experienced. This sequencing is grounded in the cognitive neuroscience of learning (Immordino-Yang & Damasio, 2007) and resonates with emerging scholarship on the role of emotion in engineering learning (Kellam et al., 2018; Lönngren et al., 2020), yet its systematic implementation at the course level has not, to our knowledge, been previously documented in the engineering education literature.

Scenario-Based Learning and Immersive Practice

Situated learning theory (Lave & Wenger, 1991) posits that knowledge acquisition cannot be divorced from its application context. This principle underlies scenario-based learning approaches, which place students in authentic professional situations requiring decision-making under constraints (Errington, 2011). In engineering education, scenarios have been employed in various forms—role-plays, simulations, design challenges—with documented benefits for professional skill development (Bédard et al., 2012). Schön's (1983) concept of the "reflective practicum"—a setting designed to help students acquire the artistry of professional practice through doing under coaching—provides the theoretical foundation for scenario-based approaches.

However, scenario-based learning in engineering education typically functions as a standalone activity rather than as one dimension of an integrated pedagogical progression. Students may participate in a simulation or role-play, but the activity is rarely systematically connected to prior narrative engagement (which establishes **why** the scenario matters) or subsequent creative production (which requires students to **apply** what the scenario taught them). Prince and Felder (2006), in their comprehensive review of inductive teaching methods, noted that the effectiveness of any single active-learning technique depends substantially on how it is sequenced within the broader instructional design—yet few published accounts provide detailed sequencing frameworks.

The NFSC model addresses this gap by positioning scenario-based learning as the **third** dimension of a four-stage progression. Scenarios in NFSC are not standalone events; they are explicitly designed to bridge the framework-equipped analytical capability developed in Week 2 and the creative production required in Week 8. This sequential integration—narrative establishes why the problem matters, framework provides the analytical tools, scenario tempers those tools through authentic practice, and creation consolidates learning through productive output—distinguishes NFSC from approaches that use scenarios in isolation.

The three scenario types employed in NFSC were designed to address progressively more complex professional competencies. The EIA debate (Week 3) requires students to adopt and defend stakeholder positions, exercising perspective-taking and argumentation skills. The Nudge design exercise (Weeks 5–6) demands creative synthesis—students must design behavioural interventions, not merely analyse existing ones. The mock environmental court (Week 7) introduces the highest level of complexity: students must navigate the tension between technical feasibility, procedural legality, and substantive justice, experiencing first-hand the ethical dimensions of engineering decision-making that are often treated as peripheral in traditional curricula (Strobel et al., 2013). Each scenario includes role cards specifying stakeholder interests and constraints, randomised event cards introducing uncertainty, and structured debriefing protocols that connect scenario experiences back to the Mismatch–Repair framework—ensuring that the experiential learning is consolidated into transferable analytical capability (Kolb, 2015).

Practitioner Research in Engineering Education

This study adopts a practitioner case study approach (Cochran-Smith & Lytle, 2009; Yin, 2018), wherein the instructor-researcher systematically documents and analyses teaching practice. Practitioner research occupies a distinct epistemological position: it does not claim generalisability through statistical inference, but rather offers "thick description" (Geertz, 1973) of situated practice that may inform other practitioners' reflection and adaptation. This approach has gained increasing legitimacy in engineering education research (Borrego & Bernhard, 2011; Case & Light, 2011), particularly for exploratory studies of novel pedagogical innovations where controlled experimental

designs are premature or infeasible. The *European Journal of Engineering Education* has published practitioner accounts of pedagogical innovation that have contributed to the broader evidence base despite small samples and context-specific designs (Edström & Kolmos, 2014; Kellam et al., 2018). As Jamieson and Lohmann (2012) argued, the engineering education community needs both "rigorous research" that establishes causal relationships and "scholarly teaching" that systematically documents innovative practice—the latter serving as the seedbed from which the former grows.

The NFSC pedagogy draws upon a unique narrative resource: seven documentary network novels authored by the course instructor, totalling approximately two million characters of non-fiction prose chronicling authentic solid waste management conflicts across China (Huang, 2025a). While the resource-intensity of this narrative base represents a significant boundary condition (discussed in §5.2), the underlying pedagogical principle—using authentic narrative to establish affective engagement before introducing analytical tools—is resource-independent and potentially transferable to other engineering disciplines (Goodson & Gill, 2011). The key contribution of this paper is not the documentary novels themselves, but rather the systematic *sequencing* of narrative engagement, framework introduction, scenario-based practice, and creative production into a coherent cognitive progression—a design principle that any instructor can adapt using locally available narrative resources (Huang, 2025b).

Methodology

Research Design

This study employs a qualitative practitioner case study design (Yin, 2018) with thematic analysis of student outputs and classroom observations (Braun & Clarke, 2006). The research is positioned as exploratory, documenting a pedagogical innovation across three iterative teaching cycles without making causal claims. The practitioner case study approach was selected for two reasons. First, controlled experimental designs were infeasible given the small cohort size ($N \approx 30$ per cycle) and the absence of a comparable parallel course section. Second, the primary research goal was to document and understand the *nature* of cognitive changes associated with NFSC—a descriptive objective well-suited to qualitative case study methodology (Stake, 1995; Merriam, 2009). The three-

cycle design (2023–2025) provided natural opportunities for iterative refinement of both the pedagogy and the data collection instruments, consistent with design-based research principles (Anderson & Shattuck, 2012).

Setting and Participants

The study was conducted in the graduate elective "Solid Waste Pollution Prevention and Control Technology" (course code 026039003, 2 credits, 32 contact hours) at [University], China, across three academic years (2023–2025, Autumn semesters). Each cycle enrolled approximately 30 master's students, with approximately 60% from environmental engineering backgrounds and 40% from cross-disciplinary backgrounds (chemistry, materials science, ecology). The course underwent three iterative revisions: v1.0 (2023, traditional lecture), v2.0 (2024, pilot NFSC), and v3.0 (2025, full NFSC implementation). Data reported in this paper draw primarily from the v2.0 and v3.0 cycles.

The NFSC Pedagogy

The NFSC pedagogy structures the eight-week course into four progressive dimensions. The Mismatch–Repair framework at the centre of the "Framework" dimension is a domain-specific analytical toolkit comprising six sub-frameworks totalling 34 diagnostic dimensions (Huang, 2025b). While the particular dimensions are tailored to solid waste management, the framework's underlying logic—using "mismatch" as an epistemological stance to reframe engineering problems from "technical failure" to "system–context misalignment"—is transferable. This approach resonates with the systems thinking tradition in engineering education (Lönngren et al., 2021) and with wicked problem frameworks (Rittel & Webber, 1973), but operationalises these abstract principles into concrete, teachable diagnostic categories.

*Narrative (Week 1). * Students pre-read excerpts from documentary network novels depicting authentic solid waste conflicts—landfill-induced land devaluation, technology-centrism in environmental impact assessments, rural waste classification failures, and community dignity claims in NIMBY disputes (Huang, 2025a). In-class discussion centres on affective engagement: "What did the villagers lose?" Analytical frameworks are deliberately withheld.

Framework (Week 2). The instructor introduces the Mismatch–Repair framework, comprising six sub-frameworks (spatial mismatch, siting justice, technological mismatch, waste time, behavioural mismatch, humanistic mismatch) totalling 34 diagnostic dimensions (Huang, 2025b). Students re-analyse Week 1's narrative excerpts using the frameworks and compare their intuitive and framework-guided analyses.

Scenario (Weeks 3–7). Three immersive scenarios are deployed: an environmental impact assessment debate (Week 3, role-playing as technical experts, regulators, and community representatives), a Nudge-based behavioural intervention design (Weeks 5–6, designing and simulating policy instruments for rural waste classification), and a mock environmental court hearing (Week 7, litigating procedural versus substantive justice in facility siting). Role cards, randomised event cards, and instructor facilitation guides structure each scenario.

Creation (Week 8). Students apply the Five-Dimensional Unified Diagnostic Matrix (Space × Technology × Behaviour × Humanism × Time) to a comprehensive case and submit output through one of three tracks: Track A (research paper analysing a solid waste problem and discussing framework limitations), Track B (engineering design for a treatment facility), or Track C (pedagogical reflection on personal learning experience).



Figure 1. The NFSC four-dimensional progression: Week 1 narrative engagement to Week 8 creative output

Figure 1. Week 1 learning page showing narrative-driven framework introduction in the NFSC pedagogy. The coloured labels indicate the dominant NFSC dimension for each week.

Data Collection and Analysis

Data sources comprised: (a) classroom observation notes recorded by the instructor immediately after each session; (b) student written outputs including in-class analysis worksheets (Week 1–2), scenario reflection memos (Weeks 3–7), and final track outputs (Week 8); and (c) a zero-configuration learning portal that tracked voluntary student interactions with pre-class materials and AI-assisted learning prompts. All student data were collected in Chinese. Quotes presented in the Findings have been translated into English by the authors and lightly edited for readability, with care taken to preserve substantive meaning. All quoted students provided written consent for the use of anonymised excerpts in research publications.

Data were analysed using thematic analysis (Braun & Clarke, 2006) following a six-phase process: familiarisation, initial coding, theme development, theme review, theme definition, and reporting. The analysis adopted a hybrid approach, combining deductive coding (guided by the research questions: attribution patterns, framework usage, output types) with inductive coding

(allowing unanticipated patterns to emerge from the data). The unit of analysis was the complete student assignment (worksheet, memo, or final output), with individual sentences and paragraphs coded for relevant content.

Two researchers independently coded a randomly selected subset of 15 student outputs (five from each track across the v2.0 and v3.0 cycles), achieving an initial simple agreement of 87%. Disagreements were resolved through discussion, after which the lead researcher applied the final coding framework to all remaining outputs. The final coding framework comprised 12 initial codes organised into three overarching themes corresponding to the research questions. The twelve codes were: moral attribution, framework-informed attribution, single-dimension framework use, multi-dimension framework use, cross-dimension analysis, terminology acquisition, research output, design output, reflective output, metacognitive awareness, emotional engagement, and framework resistance. Thematic saturation was assessed by tracking new code emergence across successive outputs; no new codes emerged after approximately 20 outputs had been coded, suggesting that the sample of ~40 student outputs from the v2.0 and v3.0 cycles was adequate for identifying the key patterns.

Classroom observations were conducted by the instructor for all eight sessions in each cycle (approximately 32 hours per cycle). Observations followed a semi-structured protocol that recorded: (a) classroom discourse patterns (student questions, discussion topics, argumentation strategies), (b) evidence of framework usage during in-class activities, and (c) instructor reflections on pedagogical adjustments for subsequent sessions. Observations were recorded as field notes within two hours of each session and served primarily as contextual data to triangulate with the student output analysis, rather than as independently analysed data sources. Member checking was conducted with five consenting students from the v3.0 cohort, who were presented with preliminary theme summaries and invited to comment on whether the researchers' interpretations resonated with their experience. Students confirmed the accuracy of the themes and provided additional contextual detail that was incorporated into the findings.

Ethical Considerations

Institutional ethics approval was obtained. Students were informed that course outputs might be used for research purposes with anonymisation; opt-out was available without academic consequence. All names and identifying details have been removed.

Findings

Three themes emerged from the analysis, corresponding to our three research questions.

Theme 1: From Moral Attribution to Framework-Informed Diagnosis

A notable shift occurred in students' causal attribution patterns between Week 5 and Week 8. During Week 5's behavioural mismatch instruction, students were asked to intuitively diagnose why a rural waste classification programme failed despite adequate infrastructure. Among $N = 28$ respondents, the most frequent attribution was "low villager quality" ($n = 11$, 39%), followed by "insufficient publicity" ($n = 6$, 21%), and "habit change is difficult" ($n = 5$, 18%). The instructor introduced the Behavioural Mismatch Four-Factor framework (cognitive barrier, convenience trap, norm deficit, feedback delay) without commenting on students' initial attributions.

By Week 8, when a structurally similar problem was presented in a different context, only $n = 2$ (7%) invoked "low quality" attribution, while $n = 7$ (25%) cited "convenience trap," $n = 6$ (21%) cited "feedback delay," and $n = 4$ (14%) cited "norm deficit." One student wrote in their Week 8 reflection: "I used to think the classification problem was about villager awareness. Now I see the waste bins are placed 300 metres from the village entrance—the 'convenience trap' dimension makes the structural design failure visible." Another reflected: "The four-factor framework gave me words for what I had only vaguely sensed. Before, I would have said 'people are lazy.' Now I understand it as a 'feedback delay'—when you sort your waste for three months and see no change in your neighbourhood, why would you continue?" A third student explicitly connected the framework to professional identity: "As an engineer, my job is not to judge people's behaviour but to design systems that make the right behaviour the easy behaviour. The framework reframed the problem from a moral one to a design one."

It would be misleading to suggest that all students experienced this attribution shift, or that the shift was equally pronounced across the cohort. Approximately four students across both cycles

continued to invoke dispositional explanations ("lack of awareness," "entrenched habits") in their Week 8 analyses despite having been introduced to the framework. These students applied the framework terminology—they could correctly identify "convenience trap" and "feedback delay" as diagnostic categories—but their analyses retained a tone of frustration with end-user behaviour rather than a focus on structural redesign. One such student wrote: "The framework helped me see the barriers more clearly, but I still think the fundamental problem is that people don't care enough about waste." This suggests that framework introduction enables, but does not guarantee, the reattribution of causality from individual to structural factors—a finding that tempers claims about the framework's transformative power and points to individual differences in cognitive flexibility or epistemic beliefs as potential moderating variables.

This attribution shift was not achieved through direct instruction about values or ethics. The instructor never told students that "blaming villagers is wrong." The shift emerged organically when students were equipped with a framework that made structural factors visible and nameable. This finding aligns with the constructive alignment principle (Biggs, 1996) but extends it: the framework itself served as the mechanism through which learning outcomes (diagnostic thinking, systems reasoning) were aligned with teaching activities (narrative reading, framework application, pre-post comparison).



Figure 2. Week 5 behavioural mismatch teaching page showing the four-factor diagnostic framework

Figure 2. Week 5 behavioural mismatch learning page, illustrating the four-factor framework that replaced students' initial intuitive attributions with structured diagnostic categories.

This shift from intuitive moral judgment ("they don't care enough") to framework-informed structural analysis ("the system makes it hard for them to care") was observed across all three teaching cycles, with the v3.0 cohort showing the most pronounced change—attributable, we suggest, to the improved scenario design in later iterations.

Theme 2: Progressive Framework Integration

Longitudinal assessment of framework usage was conducted at three time points across the eight-week cycle (Table 1). At Week 1 baseline, following initial exposure to the spatial mismatch framework, approximately half of students (n ≈ 15) were able to accurately use framework sub-dimension terminology in their written analyses. By Week 4, following exposure to four frameworks, approximately 25 students employed at least two framework dimensions when analysing a multi-dimensional case. By Week 8, approximately 12 students spontaneously integrated three or more

dimensions for cross-dimensional interaction analysis—a pattern that did not appear in earlier teaching cycles using traditional lecture methods.

Table 1. Pre-Mid-Post Framework Usage Assessment

Time Point	N	Key Metric	Students Meeting Criterion (v2.0 / v3.0)
Week 1 (Pre)	30	Accurate framework terminology	14 / 16
Week 4 (Mid)	30	≥ 2 framework dimensions	23 / 27
Week 8 (Post)	30	≥ 3 dimensions with cross-analysis	10 / 14

Note. Values are combined from the v2.0 (2024) and v3.0 (2025) cycles. Assessment was based on written student outputs scored independently by two researchers. The v3.0 cohort showed moderately higher performance at all three time points, which the study design cannot causally attribute to NFSC refinements (cohort effects, instructor experience, and concurrent curricular changes may all contribute).

Theme 3: Diversification of Creative Outputs

Across three teaching cycles, final outputs were distributed approximately as follows: Track A (research paper) 40%, Track B (engineering design) 35%, and Track C (pedagogical reflection) 25%. The Track C output—in which students analysed their own eight-week learning trajectory—proved particularly illuminating. Approximately 20% of students across all tracks demonstrated emerging metacognitive awareness in their reflections. One student in the Track C group articulated: "Before this course, I assumed solid waste problems were essentially technical. Now I recognise them as simultaneously spatial, institutional, behavioural, and temporal—the framework gave me the vocabulary to name what I previously only intuited." A second Track C student noted: "The most surprising thing was realising that I had been using the spatial mismatch framework unconsciously in Week 7 without even thinking about it. The framework had become a thinking habit, not a checklist."



Figure 3. Week 8 three-track output system with the five-dimensional diagnostic matrix

Figure 3. The three-track output system implemented in Week 8, offering research paper, engineering design, and pedagogical reflection pathways for creative production.



Figure 4. The zero-configuration learning portal providing eight-week progress tracking

Figure 4. The Learning Garden student portal displaying the eight-week progress grid with localStorage-based completion tracking.

Table 2. Thematic Coding Summary

Theme	Initial Codes	Illustrative Evidence
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| Attribution Shift | Moral/Structural; Intuitive/Framework | "people are lazy" to "feedback delay" |

| Framework Integration | Single-dim; Multi-dim; Terminology growth | 15/25/12 students at three time points |

| Output Diversification | Research; Design; Reflection; Meta-cognition | 40/35/25% track; "became a thinking habit" |

Table 3. Three-Track Output Distribution (2023–2025)

Track	Output Type	Proportion	Representative Product
A	Research paper	~40%	SWM problem analysis using diagnostic matrix
B	Engineering design	~35%	Facility process design with 3D framework justification
C	Pedagogical reflection	~25%	Meta-analysis of personal learning trajectory

Discussion

A Cognitive Progression, Not an Activity Collection

The NFSC pedagogy's core contribution is the systematic **sequencing** of narrative, framework, scenario, and creation into a coherent cognitive progression that explicitly builds motivation before introducing analytical tools. This ordering distinguishes NFSC from PBL, CDIO, and case-based teaching, all of which typically engage students through a cognitive entry point (a problem, a design brief, or a case), with motivational engagement assumed rather than pedagogically designed. This is not to suggest that these approaches lack affective dimensions entirely—many PBL implementations begin with engaging hooks, and case-based teaching often relies on the intrinsic interest of the case itself—but rather that NFSC makes the motivational-to-cognitive transition an explicit, systematically sequenced feature of the instructional design rather than an incidental byproduct.

Table 4. NFSC Compared with Established Engineering Pedagogies

| Dimension | NFSC | PBL | CDIO | Case-Based |

|-----|:--:|:--:|:--:|:--:|

| Entry Point | Affective (narrative) | Cognitive (problem) | Cognitive (conceive) | Cognitive (case) |

| Analytical Tools | Domain-specific framework | Generic problem-solving | Design process |
Case questions |

| Progressive Structure | 4-stage cognitive sequence | Iterative problem cycles | 4-stage design cycle | Standalone cases |

| Motivation Source | Story-driven engagement | Problem authenticity | Project ownership |
Case relevance |

| Creatively Output | 3-track system (paper/design/reflection) | Solution presentation |
Product/prototype | Case analysis |

| Course-Level Integration | Full 8-week sequence | Module-level | Curriculum-level | Session-level |

Transferability and Boundary Conditions

The NFSC pedagogy's transferability to other engineering disciplines, while plausible in principle, is currently speculative. The model was developed and tested exclusively within a single solid waste management course; no evidence from other disciplines, institutions, or instructors exists. Three preconditions appear necessary, though their sufficiency remains untested. First, the course must possess or develop a "narrativisable" resource base—authentic, emotionally textured accounts of professional practice. Second, the discipline's core knowledge must be abstractable into diagnostic frameworks. Third, instructors must be capable of designing and facilitating scenario-based teaching.

To illustrate how these preconditions might be met in a different discipline, consider a hypothetical adaptation to hydraulic engineering. The narrative resource could draw upon documented cases of dam construction controversies (e.g., the Three Gorges Dam resettlement, the Aral Sea ecological crisis), which contain natural narrative elements—community displacement, environmental transformation, intergenerational consequences—that could serve as affective entry points. The diagnostic framework could be adapted from existing hydraulic engineering assessment tools (e.g., multi-criteria decision analysis for water resource allocation) to create a "Water Resource Mismatch"

framework with dimensions such as spatial allocation mismatch, temporal availability mismatch, and stakeholder interest mismatch. Scenario types could include a public consultation simulation for a proposed dam project, a flood management design challenge with competing stakeholder priorities, and a transboundary water negotiation exercise. This illustrative example demonstrates the design pattern without implying that the specific implementation details of NFSC in solid waste management are directly portable. Importantly, the resource-intensity barrier—the original NFSC's reliance on a decade-long documentary novel project—remains a significant boundary condition. Short practitioner narratives or publicly available case archives could reduce this barrier, but whether they achieve comparable affective engagement is unknown.

Limitations and Reflexivity

This study carries significant limitations that must be transparently stated. The single-course, single-institution design with small cohorts limits generalisability. The absence of a controlled comparison means that observed cognitive changes cannot be causally attributed to NFSC alone—novelty effects, instructor personality, cohort composition, and concurrent coursework all represent uncontrolled confounds. The instructor's dual role as teacher and researcher introduces potential confirmation bias. The narrative resource base (Huang, 2025a) and the Mismatch–Repair framework (Huang, 2025b) are both unpublished works authored by the corresponding author—meaning that the entire pedagogical intervention depends on instructor-created resources that have not undergone independent peer review. This creates an inherent limitation for replicability: prospective adopters would need to develop their own narrative and framework resources, and the independence of the research from the pedagogy's originator is compromised. The second author's affiliation with a different institution partially mitigates this concern but does not eliminate it. These limitations position the study squarely within the tradition of exploratory practitioner research (Cochran-Smith & Lytle, 2009) rather than confirmatory scholarship. As with all case study research (Yin, 2018), the value lies not in statistical generalisation but in analytical generalisation—the extent to which the documented mechanisms (narrative-triggered engagement, framework-guided analysis, scenario-tempered application) may be transferable to other settings.

Implications for Engineering Education

The NFSC model carries four implications for engineering education research and practice.

First, it demonstrates that the "why should I care" problem—frequently cited as a challenge in engineering courses (Felder & Brent, 2003)—can be addressed not through extrinsic motivators (grades, career relevance) but through pedagogically embedded narrative engagement. This finding challenges the implicit assumption in many active-learning frameworks that motivation is either pre-existing or externally provided. NFSC suggests that motivation can be systematically *designed into* the instructional sequence through deliberate narrative placement.

Second, it suggests that domain-specific analytical frameworks may serve as more effective cognitive scaffolds than generic problem-solving strategies, particularly when introduced *after* narrative engagement has established the relevance of the analytical categories. The Behavioural Mismatch Four-Factor framework, for instance, transformed students' diagnoses not by teaching new technical content but by providing a vocabulary that made structural factors visible—a finding consistent with the "cognitive apprenticeship" model (Collins et al., 1991) and with research on the role of disciplinary language in professional identity formation (Becher & Trowler, 2001).

Third, it illustrates that a coherent four-stage progression (narrative → framework → scenario → creation) can be implemented within the temporal constraints of a standard eight-week graduate course, addressing the persistent "last mile" challenge of translating curricular ambitions into weekly instructional designs (Biggs & Tang, 2011). The eight-week NFSC sequence deliberately mirrors the temporal structure of many graduate modules, suggesting that the model may be adaptable to diverse institutional contexts without requiring fundamental timetable restructuring.

Fourth, the model's transparent acknowledgment of its limitations—explicitly documenting the resource-intensity of the narrative base and the absence of controlled comparison—exemplifies scholarly teaching (Jamieson & Lohmann, 2012) that is rigorous without being defensive. As the engineering education community continues to debate the appropriate balance between "rigorous research" and "reflective practice" (Borrego & Bernhard, 2011), NFSC offers an example of how

practitioner scholarship can generate transferable design principles while transparently documenting the conditions under which those principles were developed.

Conclusion

The NFSC pedagogy offers a theoretically grounded, practically implemented teaching model that addresses the structural gap between knowledge transmission and problem-solving capability in graduate engineering classrooms. Its distinctive contribution is the articulation of a cognitive progression—narrative → framework → scenario → creation—that systematically builds motivation, analytical tools, applied practice, and creative production in sequence. This progression distinguishes NFSC from established approaches such as PBL, CDIO, and case-based teaching, which typically begin with cognitive engagement (a problem, a design task, or a case) without an explicit motivational scaffolding phase.

The model's honest acknowledgment of its limitations—small scale, absence of controls, resource-intensity barriers—positions it as an invitation rather than a prescription. Other engineering educators are encouraged to adapt the NFSC design principles (narrative anchoring, framework scaffolding, scenario immersion, creation closure) to their own disciplinary contexts. The specific implementation must be reparameterised for each new domain, but the underlying progression logic is transferable.

Three directions for future research emerge from this exploratory work. First, quasi-experimental designs comparing NFSC with traditional lecture-based instruction in comparable courses, ideally across multiple institutions, could address the current study's most significant limitation—the absence of controlled comparison. Second, longitudinal tracking of graduates could assess whether framework-based diagnostic thinking persists beyond the course context into professional practice. Third, cross-disciplinary adaptations in hydraulic engineering, transportation engineering, and environmental management offer natural extensions for testing the model's generalisability beyond its original solid waste management domain.

The NFSC pedagogy's core insight—that building the motivational foundation before introducing analytical tools, through authentic narrative engagement, can transform students'

relationship to analytical frameworks—is straightforward in principle but demanding in implementation. Its implementation, however, requires careful sequencing, deliberate scenario design, and honest reflection on what works and what does not. We offer this account as a contribution to the growing body of practitioner scholarship in engineering education, in the hope that it may inspire both classroom experimentation and systematic investigation.

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